MAST20026 REAL ANALYSIS

Sequences: $\epsilon - M$ Convergence

5A

Learning Incomes

- Familiarity with the $\epsilon - M$ definition of convergence and the Algebra of Limit theorem.

Recall the following definition from Week 4:

Definition. *Let f_n be a sequence and let $L \in \mathbb{R}$. We say (f_n) converges to L when for every $\epsilon > 0$ there exists $M \in \mathbb{N}^+$ so that $|f_n - L| < \epsilon$ whenever $k > M$. When (f_n) converges to L we write $f_n \to L$ or $\lim_{n\to\infty} f_n = L$*

Intuitively we understand the notation $\lim_{n\to\infty} f_n = L$ to mean that for any $\epsilon > 0$, the terms of the sequence f_n eventually get within and stay within distance ϵ from L.

1. **Proving a Limit: Step -2**

   Each of the sequences below converge. Make a guess to which value of L each sequence converges to.

   (a) $(a_n) = \left(\frac{1}{\sqrt{n}} + 1\right)$

   (b) $(b_n) = \left(\frac{6}{3n^2-4}\right)$

   (c) $(c_n) = \left(\frac{(-1)^n}{2n}\right)$

   (d) $(d_n) = \left(\frac{n^2}{3n^2-4}\right)$

   (e) $(e_n) = (1,\ 1,\ 1,\ \dots)$

2. **Proving a Limit: Step -1**

   For each $\alpha \in \{a, b, c, d, e\}$ let L_α denote your guess from the previous question. Let $p(n, \epsilon)$ be the following condition

   $$|\alpha_n - L_\alpha| < \epsilon$$

   For each $\alpha \in \{a, b, c, d, e\}$ find a value $M \in \mathbb{N}^+$ so that $p(n, 0.01)$ is true whenever $n > M$.

3. **Proving a Limit: Step 0**

   For each $\alpha \in \{a, b, c, d, e\}$ find a formula for M as a function of ϵ so that $p(n, \epsilon)$ is true whenever $n > M$.

4. **Proving a Limit: Step 1**

   Write an informal proof that

   $$\lim_{n\to\infty} \frac{n^2}{3n^2 - 4} = \frac{1}{3}$$

   In your proof, you may assume algebraic manipulations work in $\mathbb{R}$ as you expect them to. Your proof should begin and end as follows:

   Let $\epsilon > 0$. Set $M =$ _____ and let $n > M$.

   [...]

   Therefore $\left|d_n - \frac{1}{3}\right| < \epsilon$ whenever $n > M$. It follows that $\lim_{n\to\infty} \frac{n^2}{3n^2 - 4} = \frac{1}{3}$.

5. **Algebra of Limits**

Recall the statement of the Algebra of Limits Theorem:

Theorem (Algebra of Limits Theorem)**.** *If $f_n \to \alpha$ and $g_n \to \beta$ are convergent sequences, then*

i. $f_n + g_n \to \alpha + \beta$;

ii. $f_n - g_n \to \alpha - \beta$;

iii. $f_n g_n \to \alpha\beta$; *and*

iv. $f_n/g_n \to \alpha/\beta$ *(provided $\beta \neq 0$ and no term of g_n is 0).*

Using the Algebra of Limits Theorem, determine the limit of each sequence. You may assume that your guesses in Question 1 are correct.

(a) $(g_n) = \left(\frac{n^2+6}{3n^2-4}\right)$

(b) $(h_n) = (3,\ 3,\ 3,\ 3,\ 3,\ 3,\ 3,\ \dots)$

6. **Comparisons**

Some of the algebra in question 1 of this tutorial took a lot of work! Looking at our definition of convergence above, there is nothing that says we must choose the least possible M, but in fact we can choose any M that works.

(a) Using the approach above, attempt to find M so that $\left|\frac{1}{n^2+6n+11} - 0\right| < \epsilon$ whenever $n > M$. Notice that while this is possible, it takes quite a lot of work and gives an answer that requires careful interpretation.

(b) Notice that for all $n \in \mathbb{N}^+$ we have

$$n^2 < n^2 + 6n + 11$$

If $|\frac{1}{n^2} - 0| < \epsilon$ is it necessarily the case that $\left|\frac{1}{n^2+6n+11} - 0\right| < \epsilon$?

(c) Find M so that $\left|\frac{1}{n^2} - 0\right| < \epsilon$ whenever $n > M$.

(d) Using M from the previous part, give a proof that

$$\lim_{n\to\infty} \frac{1}{n^2 + 6n + 11} = 0$$

(e) Where does this strategy fail if you try to use the same technique to show

$$\lim_{n\to\infty} \frac{1}{n^2 - 6n + 4} = 0?$$

Can you think of something else to do instead?